

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 3.2 ANIMAL TRANSPORT: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How does the structure of an animal's transport system determine its ability to survive and thrive, and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper? Use evidence from all twelve lessons: your own pulse-rate data, the SA:V cubes, the grasshopper, tilapia, frog and sheep or goat heart work, and the Lesson 1 and Lesson 12 models. Write your answers in the space under each prompt. Use full sentences, scientific terms and, where asked, the Claim-Evidence-Reasoning (CER) pattern. Include labelled sketches where a prompt asks for them.

Part 1: Why Transport Systems Are Needed

Eliud Kipchoge's heart rate falls from over 170 beats per minute to about 37 beats per minute soon after a race. Compare this with your own pulse recovery data after star jumps. (a) Using your SA:V cube investigation (cubes with 1 cm, 2 cm and 3 cm sides), calculate the SA:V ratio of each cube and explain what happens to the ratio as an organism gets bigger. (b) Explain why diffusion alone could not supply Kipchoge's deep muscle cells during a marathon. (c) List four jobs that a transport system does in a large, active animal.

Write your answer here:

Part 2: Open and Closed Systems – The Grasshopper

Think about the grasshopper on the school fence. (a) Describe how its circulatory system works, naming the dorsal vessel, ostia, haemocoel and haemolymph. (b) Does haemolymph carry oxygen? Explain how the grasshopper gets oxygen to its cells. (c) Why can the grasshopper survive with this system while a large, fast-moving mammal could not?

Write your answer here:

Part 3: Comparing Hearts – Tilapia, Frog and Mammal

Compare the heart and circulation of the tilapia from Lake Victoria, the frog from the Rift Valley wetlands, and a mammal such as Kipchoge. (a) Complete a table with the columns: animal, number of heart chambers, single or double circulation, and what happens to oxygenated and deoxygenated blood. (b) Explain why the tilapia's blood reaches the body with lower pressure than blood in a mammal. (c) Explain why a frog's body tissues receive only partly oxygenated blood, and why this trade-off is acceptable for a frog but not for Kipchoge.

Write your answer here:

Part 4: The Mammalian Heart and Kipchoge's Recovery

Use your sheep or goat heart observations and your pulse-rate data. (a) Describe two structural features of the mammalian heart and say what each does. (b) Explain what the sino-atrial node (SAN) does and what causes the 'lub-dub' sounds. (c) Kipchoge's heart rate is about 37 beats per minute at rest, yet each beat pumps about 135 mL, compared with about 67 mL for an average person. Explain how these figures allow his resting heart rate to be so low. Then use your pulse-rate recovery data and the Lesson 1 curves to describe how his heart rate changes after a race, and suggest what in the control of his heart rate could account for how quickly it returns to rest.

Write your answer here:

Part 5: Final Explanation (CER)

Answer the driving question using CER. Claim: how does the structure of an animal's transport system determine its ability to survive and thrive? Evidence: give at least three pieces of evidence from different lessons, naming the animals. Reasoning: explain how the evidence supports your claim. Then explain what would happen if Eliud Kipchoge had the circulatory system of a grasshopper. Finally, say how your final model differs from your Lesson 1 model.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why transport systems are needed	Links SA:V ratio, diffusion distance and the functions of transport accurately, using the cube and pulse data.	Explains that large animals need transport and names some functions, with limited use of SA:V evidence.	States that animals need transport without a clear reason or evidence.
Comparing transport systems across animals	Accurately compares grasshopper, tilapia, frog and mammal on open or closed, single or double, and mixing of blood, using correct terms.	Describes most systems correctly with a few gaps or errors.	Describes few systems, or confuses their features.
Use of evidence and CER reasoning	Gives at least three accurate pieces of evidence from different lessons and links each clearly to the claim.	Gives two or three pieces of evidence with partial reasoning.	Gives little or vague evidence; claim is not supported.
Applying ideas to Kipchoge and the grasshopper scenario	Explains fully why a grasshopper system would fail Kipchoge, covering size, pressure, oxygen carrying and recovery, using the heart rate and stroke volume data.	Explains that the system would not supply enough oxygen, with some reasons.	Gives an unsupported or incorrect prediction.
Scientific communication and model reflection	Uses scientific vocabulary precisely and clearly describes how the final model differs from the Lesson 1 model.	Uses mostly correct vocabulary and names some changes to the model.	Vocabulary is often incorrect and model changes are not described.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.